

Regression Trees (Optional) 🌳

Decision trees can be generalized to solve **regression problems**, where the goal is to predict a continuous numerical value (like a price or, in this case, an animal's **weight**) instead of a discrete class (like "Cat" or "Not Cat").

1. How Regression Trees Make Predictions

The structure of the tree (splitting on features like "Ear Shape" or "Face Shape") is built in the same way. The only difference is the value stored in the **leaf nodes**.

- **Classification Tree Leaf:** Predicts the **majority class** (e.g., "Cat") of the training examples that fall into that leaf.
- **Regression Tree Leaf:** Predicts the **average value** of the target y (e.g., the average weight) of all the training examples that fall into that leaf.

Example:

If 4 animals (with weights 7.2, 7.6, 8.4, 10.2) end up in a leaf, the prediction for that leaf becomes their average: **8.35 lbs**.

2. How Regression Trees are Trained: Reducing Variance

To build a regression tree, we need a different metric to measure "impurity." For classification, we minimized **entropy**. For regression, we minimize **variance**.

- **Entropy (Classification):** Measures the *impurity* of classes (how mixed "Cat" and "Not Cat" are).
 - **Variance (Regression):** Measures the *impurity* of numbers (how "spread out" the weights are).
 - A set of numbers like [8.8, 15, 11, 18, 20] has **high variance**.
 - A set of numbers like [7.2, 7.6, 8.4, 10.2] has **low variance**.
 - The goal is to create splits that result in nodes with the lowest possible variance.
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Finding the Best Split 📊

The algorithm's goal is to find the feature split that results in the **largest reduction in variance**. This is the direct equivalent of seeking the highest "information gain" in a classification tree.

The process at each node is:

1. **Calculate Parent Variance:** First, calculate the variance of all y values (weights) in the

current (parent) node.

- *Example:* Variance of all 10 animals = **20.51**

2. Test Each Feature Split:

For each possible feature (e.g., "Ear Shape"):

- **a.** Split the data and calculate the variance for the left and right child nodes (e.g., $\text{Variance_left} = 1.47, \text{Variance_right} = 21.87$).
- **b.** Calculate the **weighted average variance** of the children.
 - $\text{avg_variance} = (w_{\text{left}} * \text{Variance_left}) + (w_{\text{right}} * \text{Variance_right})$
 - (e.g., $(5/10 * 1.47) + (5/10 * 21.87) = 11.67$)

3. Calculate Variance Reduction:

- $\text{Variance Reduction} = \text{Parent Variance} - \text{Weighted Average Variance}$
- *Example (Ear Shape):* $20.51 - 11.67 = 8.84$
- *Example (Face Shape):* $20.51 - 19.87 = 0.64$
- *Example (Whiskers):* $20.51 - 14.29 = 6.22$

4. Decision:

- The "Ear Shape" split gives the **highest variance reduction** (8.84).
- Therefore, the algorithm chooses "Ear Shape" as the feature for this split.

This recursive process continues until a stopping criterion (like `max_depth` or minimum examples) is met.